Review of Submission 10A

First Proof Second Batch
Reviewer 1
June 4–8, 2026

1 Review

This article outlines a rather long and at times confusing path towards a proof. However, it is a significantly reasonable path. Importantly, there is significant effort that goes into various steps of the approach, making the approach seem feasible. Unfortunately, various important lemmas are missing crucial justifications, and these are explained in a very strange way. Since I cannot parse the proof structure, it is hard to make precise comments about what goes wrong. As is, this proof requires major revisions and is not close to a full proof, despite the fact that the strategy is reasonable. Here are some specific comments.

1. On a simple level I do not understand why the article keeps saying made up words like "ordinary" behind the word support, without explaining what that means, in Lemma 1. I do not also understand what "one-normal support" means. In any case, I do not understand the proof of Lemmas 1,2 but the statements seem correct.

2. I do not understand the proofs and parts of the statements of Lemmas 2 and 3, and importantly Lemma 4 as well. Large portions of these seem like just combinations taken from parts of the DKEP paper, with added jargon. It is also strangely written, and not easy to parse.

3. It is very possible that a bimodule calculation might genuinely help with proving that the support projections of the boundary pieces associated to subgraph subalgebras commute and inf down to the compacts. However, their proof of this, which follows from Lemma 4, is missing lots of details. It is important to note that this strategy is interesting, and different from the human proof.

Correctness. I think this comes closest to being a correct strategy, and is the most well developed proof of all. Unfortunately, it is not fully correct as written, but I believe it may be fixable through iterations of major revisions.

Novelty. There is one main novelty. This solution has interestingly identified a conceptually different approach towards proving that the support projections commute, using bimodule calculations, which we did not adopt in the human solution.

2 Recommendation

Require major revisions. There are multiple fixes that need to be incorporated, for instance, the usage of operator bimodules in the definition of the bidual boundary piece is missing, and additionally the bimodule computations are very confusing and unjustified. It is possible that there may be a path towards publication for this article.

3 Submission with inline comments

Problem statement and interpretation

Irreducible means not a non-trivial graph join; equivalently the complement graph Γ^c is connected. All traces are faithful normal traces. For $U \subset V$, M_U denotes the graph product over the induced subgraph and $E_U : M \to M_U$ its canonical trace-preserving expectation.

We use the notation of Ding–Kunnawalkam Elayavalli–Peterson [1]. For a finite algebra N, put $A_N = \mathbb{B}(L^2N)$ and

$$\mathcal{A}_N = (A_N^{\sharp_J})^*,$$

where $\sharp_J$ means simultaneous normality for the N- and JNJ-bimodule structures. If X is an N-boundary piece, $K_X(N)$ is DKEP's compact hereditary algebra and $q_X \in \mathcal{A}_N$ is the support of $(K_X(N)^{\sharp_J})^*$. For $X = \mathbb{K}(L^2N)$ write q_0. Set

$$\widetilde{S}_X(N) = \{T \in \mathcal{A}_N : [T, a] \in q_X \mathcal{A}_N q_X \text{ for all } a \in JNJ\}.$$

For a separable finite factor, DKEP Lemma 8.5 says that proper proximality relative to X is equivalent to the absence of an N-central state on $\widetilde{S}_X(N)$ whose restriction to N is the trace. Equivalently one may rule out N-central states whose restriction is normal, since a normal central state on a finite factor is the trace after normalization.

For a bounded vector ξ in an M-N correspondence, T_ξ denotes DKEP's coefficient operator $L^2M \to L^2N$; if η is bounded in an N-P correspondence, then

$$T_{\xi \otimes_N \eta} = T_\eta T_\xi, \qquad T_{\bar{\xi}} = T_\xi^*. \tag{1}$$

Boundary supports and coefficients

We use DKEP's ordinary-bidual realization of the normal bidual. Thus $\mathcal{A}_N$ is viewed as a von Neumann subalgebra of A_N^{**}; its unit is denoted by p_N. Products of support projections are computed inside the honest von Neumann algebra A_N^{**}, not by multiplying the non-multiplicative canonical images of concrete operators.

Lemma 1 (ordinary support bookkeeping)**.** *Let $Y \subset A_N$ be a hereditary C^*-subalgebra invariant under the multiplier actions of N and JNJ, and assume that the induced representations $N, JNJ \to M(Y)$ are faithful. Let $r_Y \in A_N^{**}$ be the ordinary open support projection of Y. Then r_Y commutes* *with p_N, and the support in $\mathcal{A}_N$ of the included corner $(Y^{\sharp_J})^*$ is $p_N r_Y$. Every contractive approximate unit of Y converges weak* in A_N^{**} to r_Y.*

If X is one of the boundary pieces used below and ρ_X denotes the ordinary support of $K_X(N)$, while r_X denotes the ordinary support of X, then

$$q_X = p_N r_X = p_N \rho_X. \tag{2}$$

Proof. Let p_ℓ and p_r be DKEP's one-normal support projections in A_N^{**}, for the N- and JNJ-module structures respectively. In the one-normal construction $p_\ell \in N^{**}$ and $p_r \in (JNJ)^{**}$. Since the concrete algebras N and JNJ commute in A_N, their bidual images commute in A_N^{**}: approximate elements of the biduals weak* by bounded nets from the concrete algebras and use separate weak* continuity of multiplication. Hence $p_\ell p_r = p_r p_\ell$.

The one-normal support paragraph preceding DKEP Remark 2.8 identifies $(A_N^{\sharp_N})^* = p_\ell A_N^{**} p_\ell$ and $(A_N^{\sharp_{JNJ}})^* = p_r A_N^{**} p_r$. DKEP Proposition 2.10 identifies the simultaneous normal bidual with their intersection inside A_N^{**}. For commuting projections e, f in a von Neumann algebra, $eRe \cap fRf = (ef)R(ef)$; therefore

$$\mathcal{A}_N = (A_N^{\sharp_J})^* = p_N A_N^{**} p_N, \qquad p_N = p_\ell p_r. \tag{3}$$

Apply the same one-normal support statement to Y for each of the two normal structures. The faithfulness hypothesis is exactly the hypothesis needed there for the multiplier actions on Y. Thus r_Y commutes with both p_ℓ and p_r, and the two one-normal biduals of Y are $r_Y p_\ell A_N^{**} p_\ell r_Y$ and $r_Y p_r A_N^{**} p_r r_Y$. Applying Proposition 2.10 to the inclusion $Y \subset A_N$ and using the elementary intersection formula once more gives

$$(Y^{\sharp_J})^* = r_Y p_N A_N^{**} p_N r_Y = (p_N r_Y)\,\mathcal{A}_N\,(p_N r_Y).$$

Thus its support projection in $\mathcal{A}_N$ is $p_N r_Y$. The weak* convergence of approximate units to r_Y is the usual open-support property in the ordinary bidual.

It remains to compare X with $K_X(N)$. For the subalgebra boundary pieces used below the faithfulness assumption is automatic: if, for example, $ae_B = 0$, then $E_B(a^*a) = 0$, hence $a = 0$, and the right action is similar; the same holds for $K_X(N)$ since $K(L^2N) \subset X \subset K_X(N)$. By the first part, the normal supports of X and $K_X(N)$ are $p_N r_X$ and $p_N \rho_X$. We show that their simultaneous-normal annihilators coincide.

Let $\phi \in A_N^{\sharp_J}$ vanish on X. DKEP Proposition 3.6 gives $K_X(N) \subset K_X^{\infty,1}(N)$, where $K_X^{\infty,1}(N)$ is the $\|\cdot\|_{\infty,1}$-closure of X. DKEP Theorem 5.6, applied with the same boundary piece on both sides, says that every $T \in K_X^{\infty,1}(N)$ is killed by every simultaneous-normal functional which vanishes on XA_NX. Since X is hereditary, $XA_NX \subset X$; hence $\phi(T) = 0$ for all $T \in K_X(N)$. The reverse annihilator inclusion is immediate from $X \subset K_X(N)$, which is part of DKEP's construction. Equal preannihilators give the same weak* closed corner in $\mathcal{A}_N$, hence the same support projection. Since q_X is by definition the support of $(K_X(N)^{\sharp_J})^*$, (2) follows. $\square$

For an expected inclusion $B \subset N$, put $H = {}_N L^2 N_B$ and

$$\mathcal{H}_B = H \otimes_B \overline{H} = {}_N L^2 N_B \otimes_B {}_B L^2 N_N.$$

Let $\mathcal{E}_B = \{T_{\eta \otimes_B \bar{\xi}} : \xi, \eta \in H_{\mathrm{b}}\}$.

Lemma 2 (subalgebra fusion)**.** *The boundary piece X_B is hereditarily generated by the positive operators s^*s, $s \in \mathcal{E}_B$, equivalently by $T_\alpha^* T_\alpha$ with α an elementary bounded tensor in $\mathcal{H}_B$. Moreover $\mathcal{H}_B \cong \overline{\mathcal{H}_B}$, and every coefficient operator T_α of a bounded vector in any Hilbert direct sum of copies of $\mathcal{H}_B$ belongs to $K_{X_B}(N)$.*

Proof. DKEP Example 5.11 applied to $H = {}_N L^2 N_B$ says that X_B is the hereditary algebra generated by $T_\xi^* T_\eta$, $\xi, \eta \in H_{\mathrm{b}}$. By (1), $T_{\eta \otimes_B \bar{\xi}} = T_{\bar{\xi}} T_\eta = T_\xi^* T_\eta$.

Let Y be the hereditary algebra generated by $\{s^*s : s \in \mathcal{E}_B\}$. Since $\mathcal{E}_B$ is self-adjoint, Y contains s^*s and ss^*. If r is the ordinary support of Y, then $(1-r)s^*s(1-r) = 0$ and $(1-r)ss^*(1-r) = 0$; hence $(1-r)s = s(1-r) = 0$. Thus $s = rsr \in rA_N^{**}r \cap A_N = Y$. So all generators $T_\xi^* T_\eta$ of X_B lie in Y, and the reverse inclusion is immediate.

The conjugation $\overline{\xi \otimes_B \bar{\eta}} \mapsto \eta \otimes_B \bar{\xi}$ gives $\overline{\mathcal{H}_B} \cong \mathcal{H}_B$. Let $\mathcal{C}$ be the elementary bounded tensors in $\mathcal{H}_B$. They are N-N cyclic, and $T_\alpha \in X_B \subset X_B A_N X_B$ for $\alpha \in \mathcal{C}$. Hence DKEP Theorem 5.10(1) applies, so $\mathcal{H}_B$ is mixing relative to $X_B \times X_B$. For a Hilbert direct sum $\bigoplus_{j \in J} \mathcal{H}_B$, take the vectors of $\mathcal{C}$ placed in one summand. Their N-N span contains the finite-support algebraic direct sum, hence is dense, and the coefficient operator of such a one-summand vector is exactly the coefficient operator above (zero against the other summands). Thus Theorem 5.10(1) holds for the whole direct sum. Theorem 5.10(3), followed by Proposition 5.9, gives $T_\alpha \in K_{X_B, X_B}(N, N) = K_{X_B}(N) A_N K_{X_B}(N) \subset K_{X_B}(N)$ for every bounded vector α. $\square$

Lemma 3 (finite fusion calculus)**.** *Let X be a boundary piece for N, with $N, JNJ \subset M(X)$. Finite Connes fusions, conjugates, and Hilbert direct sums of N-N correspondences which are mixing relative to $X \times X$ are again mixing relative to $X \times X$. Consequently, if K is any such finite-fusion correspondence and $\xi \in K$ is bounded, then $T_\xi \in K_X(N)$.*

For $U \subset V$, put $H_U = L^2 M \otimes_{M_U} L^2 M$. Let $\mathcal{T}_U$ be the linear span of all T_ξ, where ξ is a bounded vector in a finite direct sum of non-empty finite fusions of H_U and $\overline{H_U}$. Then $\mathcal{T}_U$ is a $*$-algebra, and X_{M_U} has a contractive approximate unit in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$.

Proof. For a mixing correspondence H, Theorem 5.10 says that

$$\mathcal{C}_H = \{\xi \in H_{\mathrm{b}} : T_\xi \in X A_N X\}$$

is N-N cyclic. Direct sums are handled by placing the cyclic vectors in one summand; their span contains the finite-support algebraic sum and is dense, while the coefficient condition is unchanged. If H and K are mixing and $\xi \in \mathcal{C}_H, \eta \in \mathcal{C}_K$, then $\xi \otimes_N \eta$ is bounded, such elementary tensors are cyclic for $H \otimes_N K$, and

$$T_{\xi \otimes \eta} = T_\eta T_\xi \in (X A_N X)(X A_N X) \subset X A_N X,$$

because X is hereditary. Thus the cyclic-vector criterion proves mixing for fusions. For the conjugate correspondence, the cyclic vectors are $\bar{\xi}$ and $T_{\bar{\xi}} = T_\xi^* \in X A_N X$. Iterating gives the first assertion. Applying Theorem 5.10(3) and Proposition 5.9 to any bounded vector ξ in such a correspondence gives

$$T_\xi \in K_{X,X}(N, N) = K_X(N) A_N K_X(N) \subset K_X(N).$$

The algebra property of $\mathcal{T}_U$ follows term by term from (1) with the order convention $T_\eta T_\xi = T_{\xi\otimes\eta}$: the product of two coefficient operators is the coefficient of the corresponding fused bounded vector, and the adjoint is the coefficient of the conjugate vector. Since $\mathcal{T}_U$ is defined as a linear span, this proves closure under products and adjoints. By Lemma 2, X_{M_U} is hereditarily generated by $T_\xi^* T_\xi$ for elementary tensors in H_U, and these T_ξ's belong to $\mathcal{T}_U$. If s is a finite positive sum of such generators, then $g_\varepsilon(s) = s(s+\varepsilon)^{-1}$ is in the hereditary algebra generated by the summands. Polynomials in s with zero constant term approximate $g_\varepsilon(s)^{1/2}$, so $g_\varepsilon(s)$ lies in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$. The usual net over s and $\varepsilon \downarrow 0$ is a contractive approximate unit. □

If $Z \subset U$, then $e_{M_Z} \le e_{M_U}$; hence $X_{M_Z} \subset X_{M_U}$ and $K_{X_{M_Z}}(M) \subset K_{X_{M_U}}(M)$.

Lemma 4 (intersection of boundary supports)**.** *Let $q_U = q_{X_{M_U}}$. For $U, W \subset V$, with $I = U \cap W$,*

$$0 \le q_W q_U q_W \le q_I.$$

Consequently, if $I = \cap_{i=1}^m U_i$, then

$$q_{U_m} q_{U_{m-1}} \cdots q_{U_1} q_{U_2} \cdots q_{U_{m-1}} q_{U_m} \in q_I \mathcal{A}_M q_I.$$

Proof. The graph-product bimodule decomposition and fusion rule of [2, Theorem 5.4 and Proposition 5.5] give

$$H_A \otimes_M H_B \cong \bigoplus_{C \subset A\cap B} H_C^{\oplus m(A,B,C)}. \tag{3}$$

Since $H_A \cong \overline{H_A}$, induction shows that a finite fusion whose labels are $A_1, \dots, A_n$ decomposes into a direct sum of H_C's with $C \subset A_1 \cap \cdots \cap A_n$. The implementing unitary is M-bimodular, so bounded vectors remain bounded.

Let $\mathcal{P}_R$ be the norm closure of $\{t^*t : t \in \mathcal{T}_R\}$. We claim that

$$a\, b\, c \in K_{X_{M_I}}(M) \tag{4}$$

whenever $a, c \in \mathcal{P}_W$ and $b \in \mathcal{P}_U$. It is enough to consider $a = t_1^* t_1$, $b = s^*s$, $c = t_2^* t_2$. Write t_j and s as finite sums of coefficient operators. A summand in the expansion of $t_1^* t_1 s^* s t_2^* t_2$ is

$$T_{\alpha_1}^* T_{\alpha_2} T_{\beta_1}^* T_{\beta_2} T_{\alpha_3}^* T_{\alpha_4},$$

where each α_i is a bounded vector in a finite direct sum of fusions with label W, and each β_i is similarly labelled by U. Replacing T_α^* by $T_{\bar\alpha}$ and using $T_\eta T_\xi = T_{\xi\otimes\eta}$, the displayed product is the coefficient operator of

$$\zeta := \alpha_4 \otimes \bar\alpha_3 \otimes \beta_2 \otimes \bar\beta_1 \otimes \alpha_2 \otimes \bar\alpha_1$$

(with finite direct sums distributed over the tensor products). This is a bounded vector in a finite fusion whose label list contains both W and U. By (3), it is carried by an M-bimodular unitary to a bounded vector in a direct sum of H_C's with $C \subset I$; its coefficient operator is unchanged under this identification. For each such C, Lemma 2 gives mixing, hence coefficient compactness, relative to X_{M_C}. Since $X_{M_C} \subset X_{M_I}$, the same cyclic vectors are good for X_{M_I}; the direct-sum argument in

Lemma 3 therefore puts the coefficient of the whole direct sum in $K_{X_{M_I}}(M)$. Linearity and norm closure prove (4).

Let $A = \mathbb{B}(L^2M)$, let $p = p_M$ be the unit of $\mathcal{A}_M$ in A^{**}, let r_R be the ordinary support of X_{M_R}, and let ρ_R be the ordinary support of $K_{X_{M_R}}(M)$. Lemma 1 gives

$$q_R = pr_R = p\rho_R, \qquad [p, r_R] = [p, \rho_R] = 0. \tag{5}$$

Choose contractive approximate units $h_\alpha \subset \mathcal{P}_W$ for X_{M_W} and $k_\beta \subset \mathcal{P}_U$ for X_{M_U}. In A^{**}, $h_\alpha \to r_W$ and $k_\beta \to r_U$ weak*. By (4), $h_\alpha k_\beta h_\gamma \in K_{X_{M_I}}(M) \subset \rho_I A^{**} \rho_I$. The corner $\rho_I A^{**} \rho_I$ is weak* closed, and multiplication in A^{**} is separately weak* continuous; taking the three successive limits gives

$$r_W r_U r_W \in \rho_I A^{**} \rho_I.$$

This operator is a positive contraction, hence $r_W r_U r_W \le \rho_I$. Compressing only now to the simultaneous normal bidual, (5) yields

$$q_W q_U q_W = (pr_W p)(pr_U p)(pr_W p) = p\, r_W r_U r_W\, p \le p\rho_I p = q_I.$$

The palindromic assertion follows by induction. If $R_{m-1} \le q_{\cap_{i<m} U_i}$, then

$$q_{U_m} R_{m-1} q_{U_m} \le q_{U_m} q_{\cap_{i<m} U_i} q_{U_m} \le q_{\cap_i U_i},$$

using the two-set case for the last inequality. □

From finitely many relative boundaries to the compact boundary

Lemma 5. *Let N be a separable non-amenable finite factor. Let $X_1, \dots, X_m$ be boundary pieces, $q_i = q_{X_i}$, and suppose*

$$R := q_m q_{m-1} \cdots q_1 q_2 \cdots q_{m-1} q_m \in q_0 \mathcal{A}_N q_0. \tag{6}$$

If N is properly proximal relative to each X_i, then N is properly proximal.

Proof. Assume not. DKEP Lemma 8.5 gives an N-central state ω on $\widetilde{S}(N) = \widetilde{S}_{\mathbb{K}(L^2N)}(N)$ with $\omega|_N = \tau$. Fix i and put $q = q_i$. Since q commutes with N and JNJ,

$$[q^\perp y q^\perp, a] = q^\perp [y, a] q^\perp = 0, \qquad a \in JNJ,$$

for $y \in \widetilde{S}_{X_i}(N)$; hence $q^\perp y q^\perp \in \widetilde{S}(N)$. If $\omega(q^\perp) > 0$, then $y \mapsto \omega(q^\perp y q^\perp)/\omega(q^\perp)$ is an N-central state on $\widetilde{S}_{X_i}(N)$. On N_+ it is dominated by $\omega(q^\perp)^{-1}\tau$, hence is normal; its normalization and centrality make its restriction the unique normal trace on the factor N, contradicting relative proper proximality. Thus $\omega(q_i^\perp) = 0$ for all i.

Each q_i, q_0, and the palindromic product R commute with JNJ, so they lie in $\widetilde{S}(N)$; their values below are therefore independent of any extension. Extend ω to a state on the C^*-algebra $\mathcal{A}_N$ only for the following GNS projection argument; the extension is not claimed to be N-central. In its GNS representation, the equality $\omega(q_i^\perp) = 0$ implies q_i fixes the cyclic vector. Hence

$\omega(R) = \|(q_1 q_2 \cdots q_m)\xi_\omega\|^2 = 1$. Since R is a positive contraction in $q_0\mathcal{A}_N q_0$, $R \le q_0$, and so $\omega(q_0) = 1$.

DKEP Lemma 8.5 provides the canonical N- and JNJ-bimodular u.c.p. map

$$\Theta_0 : \mathbb{B}(L^2N) \to q_0\mathcal{A}_N q_0, \qquad \Theta_0(a) = aq_0 \quad (a \in N).$$

Then $\psi(T) = \omega(\Theta_0(T))$ is an N-central state on $\mathbb{B}(L^2N)$. Since q_0 fixes the GNS cyclic vector, $\psi(a) = \omega(aq_0) = \omega(a) = \tau(a)$ for $a \in N$. This Connes hypertrace contradicts non-amenability of N. $\square$

A relative Bass–Serre paradox

Proposition 1. *Let $N = P *_B Q$ with trace-preserving expectations and $Q = B \overline{\otimes} A$. Assume that A has a trace-zero unitary a, and that P has unitaries w_1, w_2 satisfying*

$$E_B(w_i) = 0, \qquad E_B(w_1^* w_2) = 0.$$

Then N is properly proximal relative to X_B.

Proof. Let $H_P = L^2P \ominus L^2B$ and $H_Q = L^2Q \ominus L^2B$. The AFP normal form is

$$L^2N = L^2B \oplus \bigoplus_{\substack{n\ge1,\ i_j\in\{P,Q\}\\ i_j\ne i_{j+1}}} H_{i_1} \otimes_B \cdots \otimes_B H_{i_n}.$$

Let p and q project onto the summands whose first letter is P, respectively Q; then $1 = e_B + p + q$. Right multiplication by B preserves the first letter. Let $x \in P \ominus B$. On L^2B, JxJ creates a first-letter P vector. On a reduced word with first letter Q, right multiplication by x^* either appends or merges at the last letter, but the first letter remains Q and no L^2B-part is created. On a first-letter P word the first letter remains P, except that a length-one word may produce an L^2B-component. Hence

$$[p, JxJ] = (1 - e_B)JxJe_B - e_BJxJ(1 - e_B), \qquad [q, JxJ] = 0. \tag{7}$$

The symmetric formula holds with P, p and Q, q interchanged for $x \in Q \ominus B$. Since $E_B(x) = 0$, $e_BJxJe_B = 0$; hence the two terms in (7) are just right translates of e_B (up to sign), so they lie in $X_B \subset K^{\infty,1}_{X_B}(N)$.

We now translate DKEP Lemma 6.1. For a fixed operator T, the set of $x \in N$ such that both $[T, JxJ]$ and $[T, Jx^*J]$ lie in $K^{\infty,1}_{X_B}(N)$ is a von Neumann algebra; this is Lemma 6.1 applied to $B(L^2N)$ with DKEP's right-action convention $x \cdot T \cdot y = Jy^*JTJx^*J$. For $T = p$ and $T = q$ this von Neumann algebra contains B, $P \ominus B$, and $Q \ominus B$, hence contains the von Neumann algebra generated by P and Q. Thus $p, q \in S_{X_B}(N)$.

It remains, by DKEP Theorem 6.2, to rule out an N-central state φ on $S_{X_B}(N)$ with normal restriction to N. For $\eta \in qL^2N$, $w_i\eta$ is a reduced vector with first letter P. Moreover $q\,w_1^*w_2\,q = 0$: if a reduced vector begins with Q, left multiplication by the centered P-letter $w_1^*w_2$ produces a vector beginning with P. Thus

$$w_1qw_1^* + w_2qw_2^* \le p.$$

Likewise $a \in Q \ominus B$ and left multiplication by a sends every first-letter P reduced word into a first-letter Q word, so

$$apa^* \leq q.$$

Since $S_{X_B}(N)$ is an N-bimodule and φ is N-central, these inequalities give $2\varphi(q) \leq \varphi(p) \leq \varphi(q)$, hence $\varphi(p) = \varphi(q) = 0$.

The unitary $h = w_1 a$ is B-Haar: every non-zero power is an alternating reduced word in centered P- and Q-letters, so $E_B(h^n) = 0$ for $n \neq 0$. For $m \neq n$,

$$h^n e_B h^{-n} h^m e_B h^{-m} = h^n e_B h^{m-n} e_B h^{-m} = h^n E_B(h^{m-n}) e_B h^{-m} = 0.$$

The projections $h^n e_B h^{-n}$ are pairwise orthogonal and, by centrality, have the same φ-value; hence $\varphi(e_B) = 0$. This contradicts $1 = \varphi(e_B + p + q)$. By DKEP Theorem 6.2 this is precisely proper proximality relative to X_B (and, in the factor case used below, agrees with the normal-bidual criterion from Lemma 8.5). $\square$

Application to graph products

Fix $v \in V$ and put $L_v = \mathrm{Lk}_\Gamma(v)$. The graph-product reduced-word model, or [3], gives the standard amalgamated free product decomposition

$$M \cong M_{V\setminus\{v\}} *_{M_{L_v}} (M_{L_v} \overline{\otimes} M_v). \tag{8}$$

Let $C(v) = \{r \neq v : r \not\sim_\Gamma v\}$. Since Γ^c is connected, $C(v) \neq \emptyset$. If $C(v)$ contains distinct r, t, choose trace-zero unitaries u_r, u_t and set $w_1 = u_r, w_2 = u_t$. Then $w_i \notin M_{L_v}$, and the reduced-word expectation onto M_{L_v} kills w_i and $w_1^* w_2$. If $C(v) = \{r\}$, connectedness of Γ^c and $|V| \geq 3$ give $s \in L_v$ with $s \not\sim_\Gamma r$; otherwise $\{v, r\}$ would be a component of Γ^c. Set $w_1 = u_r, w_2 = u_s u_r$. Here $E_{M_{L_v}}(w_1) = 0$, $E_{M_{L_v}}(w_2) = u_s E_{M_{L_v}}(u_r) = 0$, and $w_1^* w_2 = u_r^* u_s u_r$ is a reduced word containing the outside vertex r, so its expectation onto M_{L_v} is 0. Thus in all cases

$$E_{M_{L_v}}(w_i) = 0, \qquad E_{M_{L_v}}(w_1^* w_2) = 0.$$

A trace-zero unitary in M_v supplies the unitary a in (8), so Proposition 1 shows that M is properly proximal relative to $X_{M_{L_v}}$ for every v.

Enumerate $V = \{v_1, \ldots, v_m\}$. Since no vertex belongs to its own link, $\cap_i L_{v_i} = \emptyset$. Lemma 4 puts the associated palindromic product in $q_\emptyset \mathcal{A}_M q_\emptyset$. Here $M_\emptyset = \mathbb{C}$, $e_\mathbb{C}$ is the rank-one projection onto $\mathbb{C}\hat{1}$, and DKEP Example 3.4 gives $X_{M_\emptyset} = \mathbb{K}(L^2 M)$.

Finally, CDHJKN Theorem 2.4 implies that a join-irreducible graph product with $|V| \geq 3$ and trace-zero vertex unitaries is a diffuse full factor. A diffuse amenable finite factor is the hyperfinite II_1 factor and is not full, so M is non-amenable. It is separable because the graph is finite and the vertices are separable. Lemma 5, applied to the finite family $\{X_{M_{L_v}} : v \in V\}$, proves that M is properly proximal.

References

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[3] M. Caspers and P. Fima, *Graph products of operator algebras*, J. Noncommut. Geom. **11** (2017), 367–411.

Review of Submission 10A

First Proof Second Batch
Reviewer 2
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1 Review

Correctness. The solution describes a correct approach. However, it is extremely badly written and contains unclear terminologies and notations as well as a gap in the proof of Lemma 1.

Novelty. The solution describes a novel approach that is different from the approach described in the human solution.

2 Recommendation

This requires major revisions. While the approach is correct, it would take an expert in the subject matter anywhere from 2 days to perhaps a week of work to rewrite this to fix all the issues and get to a publishable form. As is, it is not really understandable except to the limited number of experts who are highly familiar with the relevant concepts and can put in the work to fix the gaps and clarify the argument themselves.

3 Submission with inline comments

Problem statement and interpretation

Irreducible means not a non-trivial graph join; equivalently the complement graph Γ^c is connected. All traces are faithful normal traces. For $U \subset V$, M_U denotes the graph product over the induced subgraph and $E_U : M \to M_U$ its canonical trace-preserving expectation.

We use the notation of Ding–Kunnawalkam Elayavalli–Peterson [1]. For a finite algebra N, put $A_N = \mathbb{B}(L^2N)$ and

$$\mathcal{A}_N = (A_N^{\sharp_J})^*,$$

where $\sharp_J$ means simultaneous normality for the N- and JNJ-bimodule structures. If X is an N-boundary piece, $K_X(N)$ is DKEP's compact hereditary algebra and $q_X \in \mathcal{A}_N$ is the support of $(K_X(N)^{\sharp_J})^*$. For $X = \mathbb{K}(L^2N)$ write q_0. Set

> **Reviewer 2**
> The name “compact hereditary algebra” was never used in DKEP, nor does it make much sense. Though other than the name, the paragraph is fine.

$$\widetilde{S}_X(N) = \{T \in \mathcal{A}_N : [T, a] \in q_X \mathcal{A}_N q_X \text{ for all } a \in JNJ\}.$$

For a separable finite factor, DKEP Lemma 8.5 says that proper proximality relative to X is equivalent to the absence of an N-central state on $\widetilde{S}_X(N)$ whose restriction to N is the trace. Equivalently one may rule out N-central states whose restriction is normal, since a normal central state on a finite factor is the trace after normalization.

> **Reviewer 2**
> Both “normal” and “after normalization” are unnecessary in the above sentence.

For a bounded vector ξ in an M-N correspondence, T_ξ denotes DKEP's coefficient operator $L^2M \to L^2N$; if η is bounded in an N-P correspondence, then

$$T_{\xi \otimes_N \eta} = T_\eta T_\xi, \qquad T_{\bar{\xi}} = T_\xi^*. \tag{1}$$

Boundary supports and coefficients

We use DKEP's ordinary-bidual realization of the normal bidual. Thus $\mathcal{A}_N$ is viewed as a von Neumann subalgebra of A_N^{**}; its unit is denoted by p_N. Products of support projections are computed inside the honest von Neumann algebra A_N^{**}, not by multiplying the non-multiplicative canonical images of concrete operators.

Lemma 1 (ordinary support bookkeeping)**.** *Let $Y \subset A_N$ be a hereditary C^*-subalgebra invariant under the multiplier actions of N and JNJ, and assume that the induced representations $N, JNJ \to M(Y)$ are faithful. Let $r_Y \in A_N^{**}$ be the ordinary open support projection of Y. Then r_Y commutes with p_N, and the support in $\mathcal{A}_N$ of the included corner $(Y^{\sharp_J})^*$ is $p_N r_Y$. Every contractive approximate unit of Y converges weak* in A_N^{**} to r_Y.*

If X is one of the boundary pieces used below and ρ_X denotes the ordinary support of $K_X(N)$, while r_X denotes the ordinary support of X, then

$$q_X = p_N r_X = p_N \rho_X. \tag{2}$$

Proof. Let p_ℓ and p_r be DKEP's one-normal support projections in A_N^{**}, for the N- and JNJ-module structures respectively. In the one-normal construction $p_\ell \in N^{**}$ and $p_r \in (JNJ)^{**}$. Since the concrete algebras N and JNJ commute in A_N, their bidual images commute in A_N^{**}: approximate elements of the biduals weak* by bounded nets from the concrete algebras and use separate weak* continuity of multiplication. Hence $p_\ell p_r = p_r p_\ell$.

The one-normal support paragraph preceding DKEP Remark 2.8 identifies $(A_N^{\sharp_N})^* = p_\ell A_N^{**} p_\ell$ and $(A_N^{\sharp_{JNJ}})^* = p_r A_N^{**} p_r$. DKEP Proposition 2.10 identifies the simultaneous normal bidual with their intersection inside A_N^{**}. For commuting projections e, f in a von Neumann algebra, $eRe \cap fRf = (ef)R(ef)$; therefore

$$\mathcal{A}_N = (A_N^{\sharp_J})^* = p_N A_N^{**} p_N, \qquad p_N = p_\ell p_r. \tag{3}$$

Apply the same one-normal support statement to Y for each of the two normal structures. The faithfulness hypothesis is exactly the hypothesis needed there for the multiplier actions on Y. Thus r_Y commutes with both p_ℓ and p_r, and the two one-normal biduals of Y are $r_Y p_\ell A_N^{**} p_\ell r_Y$ and $r_Y p_r A_N^{**} p_r r_Y$. Applying Proposition 2.10 to the inclusion $Y \subset A_N$ and using the elementary intersection formula once more gives

$$(Y^{\sharp_J})^* = r_Y p_N A_N^{**} p_N r_Y = (p_N r_Y)\, \mathcal{A}_N\, (p_N r_Y).$$

Thus its support projection in $\mathcal{A}_N$ is $p_N r_Y$. The weak* convergence of approximate units to r_Y is the usual open-support property in the ordinary bidual.

It remains to compare X with $K_X(N)$. For the subalgebra boundary pieces used below the faithfulness assumption is automatic: if, for example, $ae_B = 0$, then $E_B(a^*a) = 0$, hence $a = 0$, and the right action is similar; the same holds for $K_X(N)$ since $K(L^2N) \subset X \subset K_X(N)$. By the first part, the normal supports of X and $K_X(N)$ are $p_N r_X$ and $p_N \rho_X$. We show that their simultaneous-normal annihilators coincide.

Let $\phi \in A_N^{\sharp_J}$ vanish on X. DKEP Proposition 3.6 gives $K_X(N) \subset K_X^{\infty,1}(N)$, where $K_X^{\infty,1}(N)$ is the $\|\cdot\|_{\infty,1}$-closure of X. DKEP Theorem 5.6, applied with the same boundary piece on both sides, says that every $T \in K_X^{\infty,1}(N)$ is killed by every simultaneous-normal functional which vanishes on XA_NX. Since X is hereditary, $XA_NX \subset X$; hence $\phi(T) = 0$ for all $T \in K_X(N)$. The reverse annihilator inclusion is immediate from $X \subset K_X(N)$, which is part of DKEP's construction. Equal preannihilators give the same weak* closed corner in $\mathcal{A}_N$, hence the same support projection. Since q_X is by definition the support of $(K_X(N)^{\sharp_J})^*$, (2) follows. □

> **Reviewer 2**
> While this lemma is correct as stated, the proof is badly written, contains a gap, and uses terminologies that have not been defined nor immediately clear without spending some efforts thinking about it. It needs to be completely rewritten.

For an expected inclusion $B \subset N$, put $H = {}_N L^2 N_B$ and

$$\mathcal{H}_B = H \otimes_B \overline{H} = {}_N L^2 N_B \otimes_B {}_B L^2 N_N.$$

Let $\mathcal{E}_B = \{T_{\eta \otimes_B \bar{\xi}} : \xi, \eta \in H_{\mathrm{b}}\}$.

Lemma 2 (subalgebra fusion)**.** *The boundary piece X_B is hereditarily generated by the positive operators s^*s, $s \in \mathcal{E}_B$, equivalently by $T_\alpha^* T_\alpha$ with α an elementary bounded tensor in $\mathcal{H}_B$. Moreover $\mathcal{H}_B \cong \overline{\mathcal{H}_B}$, and every coefficient operator T_α of a bounded vector in any Hilbert direct sum of copies of $\mathcal{H}_B$ belongs to $K_{X_B}(N)$.*

Proof. DKEP Example 5.11 applied to $H = {}_N L^2 N_B$ says that X_B is the hereditary algebra generated by $T_\xi^* T_\eta$, $\xi, \eta \in H_{\mathrm{b}}$. By (1), $T_{\eta \otimes_B \bar{\xi}} = T_{\bar{\xi}} T_\eta = T_\xi^* T_\eta$.

Let Y be the hereditary algebra generated by $\{s^*s : s \in \mathcal{E}_B\}$. Since $\mathcal{E}_B$ is self-adjoint, Y contains s^*s and ss^*. If r is the ordinary support of Y, then $(1-r)s^*s(1-r) = 0$ and $(1-r)ss^*(1-r) = 0$; hence $(1-r)s = s(1-r) = 0$. Thus $s = rsr \in rA_N^{**}r \cap A_N = Y$. So all generators $T_\xi^* T_\eta$ of X_B lie in Y, and the reverse inclusion is immediate.

The conjugation $\overline{\xi \otimes_B \bar{\eta}} \mapsto \eta \otimes_B \bar{\xi}$ gives $\overline{\mathcal{H}_B} \cong \mathcal{H}_B$. Let $\mathcal{C}$ be the elementary bounded tensors in $\mathcal{H}_B$. They are N-N cyclic, and $T_\alpha \in X_B \subset X_B A_N X_B$ for $\alpha \in \mathcal{C}$. Hence DKEP Theorem 5.10(1) applies, so $\mathcal{H}_B$ is mixing relative to $X_B \times X_B$. For a Hilbert direct sum $\bigoplus_{j\in J} \mathcal{H}_B$, take the vectors of $\mathcal{C}$ placed in one summand. Their N-N span contains the finite-support algebraic direct sum, hence is dense, and the coefficient operator of such a one-summand vector is exactly the coefficient operator above (zero against the other summands). Thus Theorem 5.10(1) holds for the whole direct sum. Theorem 5.10(3), followed by Proposition 5.9, gives $T_\alpha \in K_{X_B,X_B}(N,N) = K_{X_B}(N)A_N K_{X_B}(N) \subset K_{X_B}(N)$ for every bounded vector α. □

> **Reviewer 2**
> This needs to be rewritten and clarified, but the essential idea seems correct. It's hard to be specific about how to fix the proof though as it is badly rewritten and requires a more global rewrite.

Lemma 3 (finite fusion calculus)**.** *Let X be a boundary piece for N, with $N, JNJ \subset M(X)$. Finite Connes fusions, conjugates, and Hilbert direct sums of N-N correspondences which are mixing relative to $X \times X$ are again mixing relative to $X \times X$. Consequently, if K is any such finite-fusion correspondence and $\xi \in K$ is bounded, then $T_\xi \in K_X(N)$.*

For $U \subset V$, put $H_U = L^2M \otimes_{M_U} L^2M$. Let $\mathcal{T}_U$ be the linear span of all T_ξ, where ξ is a bounded vector in a finite direct sum of non-empty finite fusions of H_U and $\overline{H_U}$. Then $\mathcal{T}_U$ is a $*$-algebra, and X_{M_U} has a contractive approximate unit in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$.

Proof. For a mixing correspondence H, Theorem 5.10 says that

$$\mathcal{C}_H = \{\xi \in H_{\mathrm{b}} : T_\xi \in XA_NX\}$$

is N-N cyclic. Direct sums are handled by placing the cyclic vectors in one summand; their span contains the finite-support algebraic sum and is dense, while the coefficient condition is unchanged. If H and K are mixing and $\xi \in \mathcal{C}_H, \eta \in \mathcal{C}_K$, then $\xi \otimes_N \eta$ is bounded, such elementary tensors are cyclic for $H \otimes_N K$, and

$$T_{\xi\otimes\eta} = T_\eta T_\xi \in (XA_NX)(XA_NX) \subset XA_NX,$$

because X is hereditary. Thus the cyclic-vector criterion proves mixing for fusions. For the conjugate correspondence, the cyclic vectors are $\bar{\xi}$ and $T_{\bar{\xi}} = T_\xi^* \in XA_NX$. Iterating gives the first assertion. Applying Theorem 5.10(3) and Proposition 5.9 to any bounded vector ξ in such a correspondence gives

$$T_\xi \in K_{X,X}(N,N) = K_X(N)A_NK_X(N) \subset K_X(N).$$

The algebra property of $\mathcal{T}_U$ follows term by term from (1) with the order convention $T_\eta T_\xi = T_{\xi\otimes\eta}$: the product of two coefficient operators is the coefficient of the corresponding fused bounded vector, and the adjoint is the coefficient of the conjugate vector. Since $\mathcal{T}_U$ is defined as a linear span, this proves closure under products and adjoints. By Lemma 2, X_{M_U} is hereditarily generated by $T_\xi^*T_\xi$ for elementary tensors in H_U, and these T_ξ's belong to $\mathcal{T}_U$. If s is a finite positive sum of such generators, then $g_\varepsilon(s) = s(s+\varepsilon)^{-1}$ is in the hereditary algebra generated by the summands. Polynomials in s with zero constant term approximate $g_\varepsilon(s)^{1/2}$, so $g_\varepsilon(s)$ lies in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$. The usual net over s and $\varepsilon \downarrow 0$ is a contractive approximate unit. □

Reviewer 2

Similar to the previous lemma, this needs to be rewritten and clarified in a way that's hard to be specific about as it requires a more global rewrite, but the essential idea seems correct as well.

If $Z \subset U$, then $e_{M_Z} \leq e_{M_U}$; hence $X_{M_Z} \subset X_{M_U}$ and $K_{X_{M_Z}}(M) \subset K_{X_{M_U}}(M)$.

Lemma 4 (intersection of boundary supports)**.** *Let $q_U = q_{X_{M_U}}$. For $U, W \subset V$, with $I = U \cap W$,*

$$0 \leq q_W q_U q_W \leq q_I.$$

Consequently, if $I = \cap_{i=1}^m U_i$, then

$$q_{U_m} q_{U_{m-1}} \cdots q_{U_1} q_{U_2} \cdots q_{U_{m-1}} q_{U_m} \in q_I \mathcal{A}_M q_I.$$

Proof. The graph-product bimodule decomposition and fusion rule of [2, Theorem 5.4 and Proposition 5.5] give

$$H_A \otimes_M H_B \cong \bigoplus_{C \subset A \cap B} H_C^{\oplus m(A,B,C)}. \tag{3}$$

Since $H_A \cong \overline{H_A}$, induction shows that a finite fusion whose labels are $A_1, \ldots, A_n$ decomposes into a direct sum of H_C's with $C \subset A_1 \cap \cdots \cap A_n$. The implementing unitary is M-bimodular, so bounded vectors remain bounded.

Let $\mathcal{P}_R$ be the norm closure of $\{t^*t : t \in \mathcal{T}_R\}$. We claim that

$$a\, b\, c \in K_{X_{M_I}}(M) \tag{4}$$

whenever $a, c \in \mathcal{P}_W$ and $b \in \mathcal{P}_U$. It is enough to consider $a = t_1^* t_1$, $b = s^* s$, $c = t_2^* t_2$. Write t_j and s as finite sums of coefficient operators. A summand in the expansion of $t_1^* t_1 s^* s t_2^* t_2$ is

$$T_{\alpha_1}^* T_{\alpha_2} T_{\beta_1}^* T_{\beta_2} T_{\alpha_3}^* T_{\alpha_4},$$

where each α_i is a bounded vector in a finite direct sum of fusions with label W, and each β_i is similarly labelled by U. Replacing T_α^* by $T_{\bar{\alpha}}$ and using $T_\eta T_\xi = T_{\xi \otimes \eta}$, the displayed product is the coefficient operator of

$$\zeta := \alpha_4 \otimes \bar{\alpha}_3 \otimes \beta_2 \otimes \bar{\beta}_1 \otimes \alpha_2 \otimes \bar{\alpha}_1$$

(with finite direct sums distributed over the tensor products). This is a bounded vector in a finite fusion whose label list contains both W and U. By (3), it is carried by an M-bimodular unitary to a bounded vector in a direct sum of H_C's with $C \subset I$; its coefficient operator is unchanged under this identification. For each such C, Lemma 2 gives mixing, hence coefficient compactness, relative to X_{M_C}. Since $X_{M_C} \subset X_{M_I}$, the same cyclic vectors are good for X_{M_I}; the direct-sum argument in Lemma 3 therefore puts the coefficient of the whole direct sum in $K_{X_{M_I}}(M)$. Linearity and norm closure prove (4).

Let $A = \mathbb{B}(L^2 M)$, let $p = p_M$ be the unit of $\mathcal{A}_M$ in A^{**}, let r_R be the ordinary support of X_{M_R}, and let ρ_R be the ordinary support of $K_{X_{M_R}}(M)$. Lemma 1 gives

$$q_R = p r_R = p \rho_R, \qquad [p, r_R] = [p, \rho_R] = 0. \tag{5}$$

Choose contractive approximate units $h_\alpha \subset \mathcal{P}_W$ for X_{M_W} and $k_\beta \subset \mathcal{P}_U$ for X_{M_U}. In A^{**}, $h_\alpha \to r_W$ and $k_\beta \to r_U$ weak*. By (4), $h_\alpha k_\beta h_\gamma \in K_{X_{M_I}}(M) \subset \rho_I A^{**} \rho_I$. The corner $\rho_I A^{**} \rho_I$ is weak* closed, and multiplication in A^{**} is separately weak* continuous; taking the three successive limits gives

$$r_W r_U r_W \in \rho_I A^{**} \rho_I.$$

This operator is a positive contraction, hence $r_W r_U r_W \leq \rho_I$. Compressing only now to the simultaneous normal bidual, (5) yields

$$q_W q_U q_W = (p r_W p)(p r_U p)(p r_W p) = p\, r_W r_U r_W\, p \leq p \rho_I p = q_I.$$

The palindromic assertion follows by induction. If $R_{m-1} \leq q_{\cap_{i<m} U_i}$, then

$$q_{U_m} R_{m-1} q_{U_m} \leq q_{U_m} q_{\cap_{i<m} U_i} q_{U_m} \leq q_{\cap_i U_i},$$

using the two-set case for the last inequality. $\square$

> **Reviewer 2**
>
> This lemma presents a novel approach and the proof is generally correct. However, the AI seems to have a lack of understanding of the arguments needed and does not include an immediate consequence that would have simplified the later arguments significantly.

From finitely many relative boundaries to the compact boundary

Lemma 5. *Let N be a separable non-amenable finite factor. Let $X_1, \ldots, X_m$ be boundary pieces, $q_i = q_{X_i}$, and suppose*

$$R := q_m q_{m-1} \cdots q_1 q_2 \cdots q_{m-1} q_m \in q_0 \mathcal{A}_N q_0. \tag{6}$$

If N is properly proximal relative to each X_i, then N is properly proximal.

Proof. Assume not. DKEP Lemma 8.5 gives an N-central state ω on $\widetilde{S}(N) = \widetilde{S}_{\mathbb{K}(L^2 N)}(N)$ with $\omega|_N = \tau$. Fix i and put $q = q_i$. Since q commutes with N and JNJ,

$$[q^\perp y q^\perp, a] = q^\perp [y, a] q^\perp = 0, \qquad a \in JNJ,$$

for $y \in \widetilde{S}_{X_i}(N)$; hence $q^\perp y q^\perp \in \widetilde{S}(N)$. If $\omega(q^\perp) > 0$, then $y \mapsto \omega(q^\perp y q^\perp)/\omega(q^\perp)$ is an N-central state on $\widetilde{S}_{X_i}(N)$. On N_+ it is dominated by $\omega(q^\perp)^{-1}\tau$, hence is normal; its normalization and centrality make its restriction the unique normal trace on the factor N, contradicting relative proper proximality. Thus $\omega(q_i^\perp) = 0$ for all i.

Each q_i, q_0, and the palindromic product R commute with JNJ, so they lie in $\widetilde{S}(N)$; their values below are therefore independent of any extension. Extend ω to a state on the C^*-algebra $\mathcal{A}_N$ only for the following GNS projection argument; the extension is not claimed to be N-central. In its GNS representation, the equality $\omega(q_i^\perp) = 0$ implies q_i fixes the cyclic vector. Hence $\omega(R) = \|(q_1 q_2 \cdots q_m)\xi_\omega\|^2 = 1$. Since R is a positive contraction in $q_0 \mathcal{A}_N q_0$, $R \leq q_0$, and so $\omega(q_0) = 1$.

DKEP Lemma 8.5 provides the canonical N- and JNJ-bimodular u.c.p. map

$$\Theta_0 : \mathbb{B}(L^2N) \to q_0\mathcal{A}_N q_0, \qquad \Theta_0(a) = aq_0 \quad (a \in N).$$

Then $\psi(T) = \omega(\Theta_0(T))$ is an N-central state on $\mathbb{B}(L^2N)$. Since q_0 fixes the GNS cyclic vector, $\psi(a) = \omega(aq_0) = \omega(a) = \tau(a)$ for $a \in N$. This Connes hypertrace contradicts non-amenability of N. $\square$

A relative Bass–Serre paradox

Proposition 1. *Let $N = P *_B Q$ with trace-preserving expectations and $Q = B \overline{\otimes} A$. Assume that A has a trace-zero unitary a, and that P has unitaries w_1, w_2 satisfying*

$$E_B(w_i) = 0, \qquad E_B(w_1^* w_2) = 0.$$

Then N is properly proximal relative to X_B.

Proof. Let $H_P = L^2P \ominus L^2B$ and $H_Q = L^2Q \ominus L^2B$. The AFP normal form is

$$L^2N = L^2B \oplus \bigoplus_{\substack{n\geq 1,\ i_j \in \{P,Q\} \\ i_j \neq i_{j+1}}} H_{i_1} \otimes_B \cdots \otimes_B H_{i_n}.$$

Let p and q project onto the summands whose first letter is P, respectively Q; then $1 = e_B + p + q$. Right multiplication by B preserves the first letter. Let $x \in P \ominus B$. On L^2B, JxJ creates a first-letter P vector. On a reduced word with first letter Q, right multiplication by x^* either appends or merges at the last letter, but the first letter remains Q and no L^2B-part is created. On a first-letter P word the first letter remains P, except that a length-one word may produce an L^2B-component. Hence

$$[p, JxJ] = (1 - e_B)JxJe_B - e_BJxJ(1 - e_B), \qquad [q, JxJ] = 0. \tag{7}$$

The symmetric formula holds with P, p and Q, q interchanged for $x \in Q \ominus B$. Since $E_B(x) = 0$, $e_BJxJe_B = 0$; hence the two terms in (7) are just right translates of e_B (up to sign), so they lie in $X_B \subset K^{\infty,1}_{X_B}(N)$.

We now translate DKEP Lemma 6.1. For a fixed operator T, the set of $x \in N$ such that both $[T, JxJ]$ and $[T, Jx^*J]$ lie in $K^{\infty,1}_{X_B}(N)$ is a von Neumann algebra; this is Lemma 6.1 applied to $B(L^2N)$ with DKEP's right-action convention $x \cdot T \cdot y = Jy^*JTJx^*J$. For $T = p$ and $T = q$ this von Neumann algebra contains B, $P \ominus B$, and $Q \ominus B$, hence contains the von Neumann algebra generated by P and Q. Thus $p, q \in S_{X_B}(N)$.

It remains, by DKEP Theorem 6.2, to rule out an N-central state φ on $S_{X_B}(N)$ with normal restriction to N. For $\eta \in qL^2N$, $w_i\eta$ is a reduced vector with first letter P. Moreover $q\, w_1^* w_2\, q = 0$: if a reduced vector begins with Q, left multiplication by the centered P-letter $w_1^* w_2$ produces a vector beginning with P. Thus

$$w_1 q w_1^* + w_2 q w_2^* \leq p.$$

Likewise $a \in Q \ominus B$ and left multiplication by a sends every first-letter P reduced word into a first-letter Q word, so

$$apa^* \leq q.$$

Since $S_{X_B}(N)$ is an N-bimodule and φ is N-central, these inequalities give $2\varphi(q) \leq \varphi(p) \leq \varphi(q)$, hence $\varphi(p) = \varphi(q) = 0$.

The unitary $h = w_1 a$ is B-Haar: every non-zero power is an alternating reduced word in centered P- and Q-letters, so $E_B(h^n) = 0$ for $n \neq 0$. For $m \neq n$,

$$h^n e_B h^{-n} h^m e_B h^{-m} = h^n e_B h^{m-n} e_B h^{-m} = h^n E_B(h^{m-n}) e_B h^{-m} = 0.$$

The projections $h^n e_B h^{-n}$ are pairwise orthogonal and, by centrality, have the same φ-value; hence $\varphi(e_B) = 0$. This contradicts $1 = \varphi(e_B + p + q)$. By DKEP Theorem 6.2 this is precisely proper proximality relative to X_B (and, in the factor case used below, agrees with the normal-bidual criterion from Lemma 8.5). $\square$

Application to graph products

Fix $v \in V$ and put $L_v = \mathrm{Lk}_\Gamma(v)$. The graph-product reduced-word model, or [3], gives the standard amalgamated free product decomposition

$$M \cong M_{V\setminus\{v\}} *_{M_{L_v}} (M_{L_v} \overline{\otimes} M_v). \tag{8}$$

Let $C(v) = \{r \neq v : r \not\sim_\Gamma v\}$. Since Γ^c is connected, $C(v) \neq \emptyset$. If $C(v)$ contains distinct r, t, choose trace-zero unitaries u_r, u_t and set $w_1 = u_r, w_2 = u_t$. Then $w_i \notin M_{L_v}$, and the reduced-word expectation onto M_{L_v} kills w_i and $w_1^* w_2$. If $C(v) = \{r\}$, connectedness of Γ^c and $|V| \geq 3$ give $s \in L_v$ with $s \not\sim_\Gamma r$; otherwise $\{v, r\}$ would be a component of Γ^c. Set $w_1 = u_r, w_2 = u_s u_r$. Here $E_{M_{L_v}}(w_1) = 0$, $E_{M_{L_v}}(w_2) = u_s E_{M_{L_v}}(u_r) = 0$, and $w_1^* w_2 = u_r^* u_s u_r$ is a reduced word containing the outside vertex r, so its expectation onto M_{L_v} is 0. Thus in all cases

$$E_{M_{L_v}}(w_i) = 0, \qquad E_{M_{L_v}}(w_1^* w_2) = 0.$$

A trace-zero unitary in M_v supplies the unitary a in (8), so Proposition 1 shows that M is properly proximal relative to $X_{M_{L_v}}$ for every v.

Enumerate $V = \{v_1, \ldots, v_m\}$. Since no vertex belongs to its own link, $\cap_i L_{v_i} = \emptyset$. Lemma 4 puts the associated palindromic product in $q_\emptyset \mathcal{A}_M q_\emptyset$. Here $M_\emptyset = \mathbb{C}$, $e_\mathbb{C}$ is the rank-one projection onto $\mathbb{C}\hat{1}$, and DKEP Example 3.4 gives $X_{M_\emptyset} = \mathbb{K}(L^2 M)$.

Finally, CDHJKN Theorem 2.4 implies that a join-irreducible graph product with $|V| \geq 3$ and trace-zero vertex unitaries is a diffuse full factor. A diffuse amenable finite factor is the hyperfinite II_1 factor and is not full, so M is non-amenable. It is separable because the graph is finite and the vertices are separable. Lemma 5, applied to the finite family $\{X_{M_{L_v}} : v \in V\}$, proves that M is properly proximal.

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Review of Submission 10A

First Proof Second Batch
Reviewer 3
June 4–8, 2026

1 Review

Editorial Note: This additional review includes comments that were volunteered by a reviewer. The editors did not ask the reviewer to reach a decision, given that the other reports were sufficient, but have decided to release this report as part of the public record in the belief that the reviewer comments can help further indicate where any errors lie.

Correctness.

Novelty.

2 Recommendation

3 Submission with inline comments

Problem statement and interpretation

Irreducible means not a non-trivial graph join; equivalently the complement graph Γ^c is connected. All traces are faithful normal traces. For $U \subset V$, M_U denotes the graph product over the induced subgraph and $E_U : M \to M_U$ its canonical trace-preserving expectation.

We use the notation of Ding–Kunnawalkam Elayavalli–Peterson [1]. For a finite algebra N, put $A_N = \mathbb{B}(L^2N)$ and

$$\mathcal{A}_N = (A_N^{\sharp_J})^*,$$

where $\sharp_J$ means simultaneous normality for the N- and JNJ-bimodule structures. If X is an N-boundary piece, $K_X(N)$ is DKEP's compact hereditary algebra and $q_X \in \mathcal{A}_N$ is the support of $(K_X(N)^{\sharp_J})^*$. For $X = \mathbb{K}(L^2N)$ write q_0. Set

$$\widetilde{S}_X(N) = \{T \in \mathcal{A}_N : [T, a] \in q_X \mathcal{A}_N q_X \text{ for all } a \in JNJ\}.$$

For a separable finite factor, DKEP Lemma 8.5 says that proper proximality relative to X is equivalent to the absence of an N-central state on $\widetilde{S}_X(N)$ whose restriction to N is the trace. Equivalently one may rule out N-central states whose restriction is normal, since a normal central state on a finite factor is the trace after normalization.

For a bounded vector ξ in an M-N correspondence, T_ξ denotes DKEP's coefficient operator $L^2M \to L^2N$; if η is bounded in an N-P correspondence, then

$$T_{\xi\otimes_N\eta} = T_\eta T_\xi, \qquad T_{\bar{\xi}} = T_\xi^*. \tag{1}$$

Boundary supports and coefficients

We use DKEP's ordinary-bidual realization of the normal bidual. Thus $\mathcal{A}_N$ is viewed as a von Neumann subalgebra of A_N^{**}; its unit is denoted by p_N. Products of support projections are computed inside the honest von Neumann algebra A_N^{**}, not by multiplying the non-multiplicative canonical images of concrete operators.

Lemma 1 (ordinary support bookkeeping)**.** *Let $Y \subset A_N$ be a hereditary C^*-subalgebra invariant under the multiplier actions of N and JNJ, and assume that the induced representations $N, JNJ \to M(Y)$ are faithful. Let $r_Y \in A_N^{**}$ be the ordinary open support projection of Y. Then r_Y commutes with p_N, and the support in $\mathcal{A}_N$ of the included corner $(Y^{\sharp_J})^*$ is $p_N r_Y$. Every contractive approximate unit of Y converges weak* in A_N^{**} to r_Y.*

If X is one of the boundary pieces used below and ρ_X denotes the ordinary support of $K_X(N)$, while r_X denotes the ordinary support of X, then

$$q_X = p_N r_X = p_N \rho_X. \tag{2}$$

Proof. Let p_ℓ and p_r be DKEP's one-normal support projections in A_N^{**}, for the N- and JNJ-module structures respectively. In the one-normal construction $p_\ell \in N^{**}$ and $p_r \in (JNJ)^{**}$. Since the concrete algebras N and JNJ commute in A_N, their bidual images commute in A_N^{**}: approximate elements of the biduals weak* by bounded nets from the concrete algebras and use separate weak* continuity of multiplication. Hence $p_\ell p_r = p_r p_\ell$.

The one-normal support paragraph preceding DKEP Remark 2.8 identifies $(A_N^{\sharp_N})^* = p_\ell A_N^{**} p_\ell$ and $(A_N^{\sharp_{JNJ}})^* = p_r A_N^{**} p_r$. DKEP Proposition 2.10 identifies the simultaneous normal bidual with their intersection inside A_N^{**}. For commuting projections e, f in a von Neumann algebra, $eRe \cap fRf = (ef)R(ef)$; therefore

$$\mathcal{A}_N = (A_N^{\sharp_J})^* = p_N A_N^{**} p_N, \qquad p_N = p_\ell p_r. \tag{3}$$

Apply the same one-normal support statement to Y for each of the two normal structures. The faithfulness hypothesis is exactly the hypothesis needed there for the multiplier actions on Y. Thus r_Y commutes with both p_ℓ and p_r, and the two one-normal biduals of Y are $r_Y p_\ell A_N^{**} p_\ell r_Y$ and $r_Y p_r A_N^{**} p_r r_Y$.

> **Reviewer 3**
>
> The proof that r_Y commutes with p_l and p_r is wrong.

Applying Proposition 2.10 to the inclusion $Y \subset A_N$ and using the elementary intersection formula once more gives

$$(Y^{\sharp_J})^* = r_Y p_N A_N^{**} p_N r_Y = (p_N r_Y)\, \mathcal{A}_N\, (p_N r_Y).$$

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It is not clear what 'elementary intersection formula' means

Thus its support projection in $\mathcal{A}_N$ is $p_N r_Y$. The weak* convergence of approximate units to r_Y is the usual open-support property in the ordinary bidual.

It remains to compare X with $K_X(N)$. For the subalgebra boundary pieces used below the faithfulness assumption is automatic: if, for example, $ae_B = 0$, then $E_B(a^*a) = 0$, hence $a = 0$, and the right action is similar; the same holds for $K_X(N)$ since $K(L^2N) \subset X \subset K_X(N)$. By the first part, the normal supports of X and $K_X(N)$ are $p_N r_X$ and $p_N \rho_X$. We show that their simultaneous-normal annihilators coincide.

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The definition of $K_X(N)$ is unclear.

Let $\phi \in A_N^{\sharp_J}$ vanish on X. DKEP Proposition 3.6 gives $K_X(N) \subset K_X^{\infty,1}(N)$, where $K_X^{\infty,1}(N)$ is the $\|\cdot\|_{\infty,1}$-closure of X. DKEP Theorem 5.6, applied with the same boundary piece on both sides, says that every $T \in K_X^{\infty,1}(N)$ is killed by every simultaneous-normal functional which vanishes on XA_NX. Since X is hereditary, $XA_NX \subset X$; hence $\phi(T) = 0$ for all $T \in K_X(N)$. The reverse annihilator inclusion is immediate from $X \subset K_X(N)$, which is part of DKEP's construction. Equal preannihilators give the same weak* closed corner in $\mathcal{A}_N$, hence the same support projection. Since q_X is by definition the support of $(K_X(N)^{\sharp_J})^*$, (2) follows. □

For an expected inclusion $B \subset N$, put $H = {}_N L^2 N_B$ and

$$\mathcal{H}_B = H \otimes_B \overline{H} = {}_N L^2 N_B \otimes_B {}_B L^2 N_N.$$

Let $\mathcal{E}_B = \{T_{\eta \otimes_B \bar{\xi}} : \xi, \eta \in H_{\mathrm{b}}\}$.

Lemma 2 (subalgebra fusion)**.** *The boundary piece X_B is hereditarily generated by the positive operators s^*s, $s \in \mathcal{E}_B$, equivalently by $T_\alpha^* T_\alpha$ with α an elementary bounded tensor in $\mathcal{H}_B$. Moreover $\mathcal{H}_B \cong \overline{\mathcal{H}_B}$, and every coefficient operator T_α of a bounded vector in any Hilbert direct sum of copies of $\mathcal{H}_B$ belongs to $K_{X_B}(N)$.*

Proof. DKEP Example 5.11 applied to $H = {}_N L^2 N_B$ says that X_B is the hereditary algebra generated by $T_\xi^* T_\eta$, $\xi, \eta \in H_{\mathrm{b}}$. By (1), $T_{\eta \otimes_B \bar{\xi}} = T_{\bar{\xi}} T_\eta = T_\xi^* T_\eta$.

Let Y be the hereditary algebra generated by $\{s^*s : s \in \mathcal{E}_B\}$. Since $\mathcal{E}_B$ is self-adjoint, Y contains s^*s and ss^*. If r is the ordinary support of Y, then $(1-r)s^*s(1-r) = 0$ and $(1-r)ss^*(1-r) = 0$; hence $(1-r)s = s(1-r) = 0$. Thus $s = rsr \in rA_N^{**}r \cap A_N = Y$. So all generators $T_\xi^* T_\eta$ of X_B lie in Y, and the reverse inclusion is immediate.

The conjugation $\overline{\xi \otimes_B \bar{\eta}} \mapsto \eta \otimes_B \bar{\xi}$ gives $\overline{\mathcal{H}_B} \cong \mathcal{H}_B$. Let $\mathcal{C}$ be the elementary bounded tensors in $\mathcal{H}_B$. They are N-N cyclic, and $T_\alpha \in X_B \subset X_B A_N X_B$ for $\alpha \in \mathcal{C}$. Hence DKEP Theorem 5.10(1) applies, so $\mathcal{H}_B$ is mixing relative to $X_B \times X_B$.

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Theorem 5.10 is about operator bimodule, but here $\mathcal{H}_B$ is a Hilbert space bimodule.

For a Hilbert direct sum $\bigoplus_{j\in J}\mathcal{H}_B$, take the vectors of $\mathcal{C}$ placed in one summand. Their N-N span contains the finite-support algebraic direct sum, hence is dense, and the coefficient operator of such a one-summand vector is exactly the coefficient operator above (zero against the other summands). Thus Theorem 5.10(1) holds for the whole direct sum. Theorem 5.10(3), followed by Proposition 5.9, gives $T_\alpha \in K_{X_B,X_B}(N,N) = K_{X_B}(N)A_N K_{X_B}(N) \subset K_{X_B}(N)$ for every bounded vector α. □

Lemma 3 (finite fusion calculus)**.** *Let X be a boundary piece for N, with $N, JNJ \subset M(X)$. Finite Connes fusions, conjugates, and Hilbert direct sums of N-N correspondences which are mixing relative to $X\times X$ are again mixing relative to $X \times X$. Consequently, if K is any such finite-fusion correspondence and $\xi \in K$ is bounded, then $T_\xi \in K_X(N)$.*

For $U \subset V$, put $H_U = L^2M \otimes_{M_U} L^2M$. Let $\mathcal{T}_U$ be the linear span of all T_ξ, where ξ is a bounded vector in a finite direct sum of non-empty finite fusions of H_U and $\overline{H_U}$. Then $\mathcal{T}_U$ is a $*$-algebra, and X_{M_U} has a contractive approximate unit in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$.

Proof.

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This citation does not mention the paper.

For a mixing correspondence H, Theorem 5.10 says that

$$\mathcal{C}_H = \{\xi \in H_{\mathrm{b}} : T_\xi \in XA_NX\}$$

is N-N cyclic. Direct sums are handled by placing the cyclic vectors in one summand; their span contains the finite-support algebraic sum and is dense, while the coefficient condition is unchanged. If H and K are mixing and $\xi \in \mathcal{C}_H, \eta \in \mathcal{C}_K$, then $\xi \otimes_N \eta$ is bounded, such elementary tensors are cyclic for $H \otimes_N K$, and

$$T_{\xi\otimes\eta} = T_\eta T_\xi \in (XA_NX)(XA_NX) \subset XA_NX,$$

because X is hereditary. Thus the cyclic-vector criterion proves mixing for fusions. For the conjugate correspondence, the cyclic vectors are $\bar{\xi}$ and $T_{\bar{\xi}} = T_\xi^* \in XA_NX$. Iterating gives the first assertion. Applying Theorem 5.10(3) and Proposition 5.9 to any bounded vector ξ in such a correspondence gives

$$T_\xi \in K_{X,X}(N,N) = K_X(N)A_NK_X(N) \subset K_X(N).$$

The algebra property of $\mathcal{T}_U$ follows term by term from (1) with the order convention $T_\eta T_\xi = T_{\xi\otimes\eta}$: the product of two coefficient operators is the coefficient of the corresponding fused bounded vector, and the adjoint is the coefficient of the conjugate vector. Since $\mathcal{T}_U$ is defined as a linear span, this proves closure under products and adjoints. By Lemma 2, X_{M_U} is hereditarily generated by $T_\xi^*T_\xi$ for elementary tensors in H_U, and these T_ξ's belong to $\mathcal{T}_U$. If s is a finite positive sum of such generators, then $g_\varepsilon(s) = s(s+\varepsilon)^{-1}$ is in the hereditary algebra generated by the summands. Polynomials in s with zero constant term approximate $g_\varepsilon(s)^{1/2}$, so $g_\varepsilon(s)$ lies in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$. The usual net over s and $\varepsilon \downarrow 0$ is a contractive approximate unit. □

If $Z \subset U$, then $e_{M_Z} \leq e_{M_U}$; hence $X_{M_Z} \subset X_{M_U}$ and $K_{X_{M_Z}}(M) \subset K_{X_{M_U}}(M)$.

Lemma 4 (intersection of boundary supports)**.** *Let $q_U = q_{X_{M_U}}$. For $U, W \subset V$, with $I = U \cap W$,*

$$0 \leq q_W q_U q_W \leq q_I.$$

Consequently, if $I = \cap_{i=1}^{m} U_i$, then

$$q_{U_m} q_{U_{m-1}} \cdots q_{U_1} q_{U_2} \cdots q_{U_{m-1}} q_{U_m} \in q_I \mathcal{A}_M q_I.$$

> **Reviewer 3**
>
> This lemma in fact implies that $[q_W, q_U] = 0$ which should significantly simply the later proofs. However, it seems that the writer did not realize this simplification.

Proof. The graph-product bimodule decomposition and fusion rule of [2, Theorem 5.4 and Proposition 5.5] give

$$H_A \otimes_M H_B \cong \bigoplus_{C \subset A \cap B} H_C^{\oplus m(A,B,C)}. \tag{3}$$

> **Reviewer 3**
>
> It is not clear what are A and B.

Since $H_A \cong \overline{H_A}$, induction shows that a finite fusion whose labels are $A_1, \ldots, A_n$ decomposes into a direct sum of H_C's with $C \subset A_1 \cap \cdots \cap A_n$. The implementing unitary is M-bimodular, so bounded vectors remain bounded.

Let $\mathcal{P}_R$ be the norm closure of $\{t^*t : t \in \mathcal{T}_R\}$. We claim that

$$a\, b\, c \in K_{X_{M_I}}(M) \tag{4}$$

whenever $a, c \in \mathcal{P}_W$ and $b \in \mathcal{P}_U$. It is enough to consider $a = t_1^* t_1$, $b = s^* s$, $c = t_2^* t_2$. Write t_j and s as finite sums of coefficient operators. A summand in the expansion of $t_1^* t_1 s^* s t_2^* t_2$ is

$$T_{\alpha_1}^* T_{\alpha_2} T_{\beta_1}^* T_{\beta_2} T_{\alpha_3}^* T_{\alpha_4},$$

where each α_i is a bounded vector in a finite direct sum of fusions with label W, and each β_i is similarly labelled by U. Replacing T_α^* by $T_{\bar{\alpha}}$ and using $T_\eta T_\xi = T_{\xi \otimes \eta}$, the displayed product is the coefficient operator of

$$\zeta := \alpha_4 \otimes \bar{\alpha}_3 \otimes \beta_2 \otimes \bar{\beta}_1 \otimes \alpha_2 \otimes \bar{\alpha}_1$$

(with finite direct sums distributed over the tensor products). This is a bounded vector in a finite fusion whose label list contains both W and U. By (3), it is carried by an M-bimodular unitary to a bounded vector in a direct sum of H_C's with $C \subset I$; its coefficient operator is unchanged under this identification. For each such C, Lemma 2 gives mixing, hence coefficient compactness, relative to X_{M_C}. Since $X_{M_C} \subset X_{M_I}$, the same cyclic vectors are good for X_{M_I}; the direct-sum argument in Lemma 3 therefore puts the coefficient of the whole direct sum in $K_{X_{M_I}}(M)$. Linearity and norm closure prove (4).

Let $A = \mathbb{B}(L^2M)$, let $p = p_M$ be the unit of $\mathcal{A}_M$ in A^{**}, let r_R be the ordinary support of X_{M_R}, and let ρ_R be the ordinary support of $K_{X_{M_R}}(M)$. Lemma 1 gives

$$q_R = pr_R = p\rho_R, \qquad [p, r_R] = [p, \rho_R] = 0. \tag{5}$$

Choose contractive approximate units $h_\alpha \subset \mathcal{P}_W$ for X_{M_W} and $k_\beta \subset \mathcal{P}_U$ for X_{M_U}. In A^{**}, $h_\alpha \to r_W$ and $k_\beta \to r_U$ weak*. By (4), $h_\alpha k_\beta h_\gamma \in K_{X_{M_I}}(M) \subset \rho_I A^{**} \rho_I$. The corner $\rho_I A^{**} \rho_I$ is weak* closed, and multiplication in A^{**} is separately weak* continuous; taking the three successive limits gives

$$r_W r_U r_W \in \rho_I A^{**} \rho_I.$$

This operator is a positive contraction, hence $r_W r_U r_W \leq \rho_I$. Compressing only now to the simultaneous normal bidual, (5) yields

$$q_W q_U q_W = (pr_W p)(pr_U p)(pr_W p) = p\, r_W r_U r_W\, p \leq p\rho_I p = q_I.$$

The palindromic assertion follows by induction. If $R_{m-1} \leq q_{\cap_{i<m} U_i}$, then

$$q_{U_m} R_{m-1} q_{U_m} \leq q_{U_m} q_{\cap_{i<m} U_i} q_{U_m} \leq q_{\cap_i U_i},$$

using the two-set case for the last inequality. □

From finitely many relative boundaries to the compact boundary

Lemma 5. *Let N be a separable non-amenable finite factor. Let $X_1, \ldots, X_m$ be boundary pieces, $q_i = q_{X_i}$, and suppose*

$$R := q_m q_{m-1} \cdots q_1 q_2 \cdots q_{m-1} q_m \in q_0 \mathcal{A}_N q_0. \tag{6}$$

If N is properly proximal relative to each X_i, then N is properly proximal.

Proof. Assume not. DKEP Lemma 8.5 gives an N-central state ω on $\widetilde{S}(N) = \widetilde{S}_{\mathbb{K}(L^2N)}(N)$ with $\omega|_N = \tau$. Fix i and put $q = q_i$. Since q commutes with N and JNJ,

$$[q^\perp y q^\perp, a] = q^\perp [y, a] q^\perp = 0, \qquad a \in JNJ,$$

for $y \in \widetilde{S}_{X_i}(N)$; hence $q^\perp y q^\perp \in \widetilde{S}(N)$. If $\omega(q^\perp) > 0$, then $y \mapsto \omega(q^\perp y q^\perp)/\omega(q^\perp)$ is an N-central state on $\widetilde{S}_{X_i}(N)$. On N_+ it is dominated by $\omega(q^\perp)^{-1}\tau$, hence is normal; its normalization and centrality make its restriction the unique normal trace on the factor N, contradicting relative proper proximality. Thus $\omega(q_i^\perp) = 0$ for all i.

Each q_i, q_0, and the palindromic product R commute with JNJ, so they lie in $\widetilde{S}(N)$; their values below are therefore independent of any extension. Extend ω to a state on the C^*-algebra $\mathcal{A}_N$ only for the following GNS projection argument; the extension is not claimed to be N-central. In its GNS representation, the equality $\omega(q_i^\perp) = 0$ implies q_i fixes the cyclic vector. Hence $\omega(R) = \|(q_1 q_2 \cdots q_m)\xi_\omega\|^2 = 1$. Since R is a positive contraction in $q_0 \mathcal{A}_N q_0$, $R \leq q_0$, and so $\omega(q_0) = 1$.

DKEP Lemma 8.5 provides the canonical N- and JNJ-bimodular u.c.p. map

$$\Theta_0 : \mathbb{B}(L^2N) \to q_0\mathcal{A}_N q_0, \qquad \Theta_0(a) = aq_0 \quad (a \in N).$$

Then $\psi(T) = \omega(\Theta_0(T))$ is an N-central state on $\mathbb{B}(L^2N)$. Since q_0 fixes the GNS cyclic vector, $\psi(a) = \omega(aq_0) = \omega(a) = \tau(a)$ for $a \in N$. This Connes hypertrace contradicts non-amenability of N. □

A relative Bass–Serre paradox

Proposition 1. *Let $N = P *_B Q$ with trace-preserving expectations and $Q = B \overline{\otimes} A$. Assume that A has a trace-zero unitary a, and that P has unitaries w_1, w_2 satisfying*

$$E_B(w_i) = 0, \qquad E_B(w_1^* w_2) = 0.$$

Then N is properly proximal relative to X_B.

Proof. Let $H_P = L^2P \ominus L^2B$ and $H_Q = L^2Q \ominus L^2B$. The AFP normal form is

$$L^2N = L^2B \oplus \bigoplus_{\substack{n\geq 1,\ i_j\in\{P,Q\}\\ i_j\neq i_{j+1}}} H_{i_1} \otimes_B \cdots \otimes_B H_{i_n}.$$

Let p and q project onto the summands whose first letter is P, respectively Q; then $1 = e_B + p + q$. Right multiplication by B preserves the first letter. Let $x \in P \ominus B$. On L^2B, JxJ creates a first-letter P vector. On a reduced word with first letter Q, right multiplication by x^* either appends or merges at the last letter, but the first letter remains Q and no L^2B-part is created. On a first-letter P word the first letter remains P, except that a length-one word may produce an L^2B-component. Hence

$$[p, JxJ] = (1 - e_B)JxJe_B - e_BJxJ(1 - e_B), \qquad [q, JxJ] = 0. \tag{7}$$

The symmetric formula holds with P, p and Q, q interchanged for $x \in Q \ominus B$. Since $E_B(x) = 0$, $e_BJxJe_B = 0$; hence the two terms in (7) are just right translates of e_B (up to sign), so they lie in $X_B \subset K^{\infty,1}_{X_B}(N)$.

We now translate DKEP Lemma 6.1. For a fixed operator T, the set of $x \in N$ such that both $[T, JxJ]$ and $[T, Jx^*J]$ lie in $K^{\infty,1}_{X_B}(N)$ is a von Neumann algebra; this is Lemma 6.1 applied to $B(L^2N)$ with DKEP's right-action convention $x \cdot T \cdot y = Jy^*JTJx^*J$. For $T = p$ and $T = q$ this von Neumann algebra contains B, $P \ominus B$, and $Q \ominus B$, hence contains the von Neumann algebra generated by P and Q. Thus $p, q \in S_{X_B}(N)$.

It remains, by DKEP Theorem 6.2, to rule out an N-central state φ on $S_{X_B}(N)$ with normal restriction to N. For $\eta \in qL^2N$, $w_i\eta$ is a reduced vector with first letter P. Moreover $q\, w_1^*w_2\, q = 0$: if a reduced vector begins with Q, left multiplication by the centered P-letter $w_1^*w_2$ produces a vector beginning with P. Thus

$$w_1qw_1^* + w_2qw_2^* \leq p.$$

Likewise $a \in Q \ominus B$ and left multiplication by a sends every first-letter P reduced word into a first-letter Q word, so

$$apa^* \leq q.$$

Since $S_{X_B}(N)$ is an N-bimodule and φ is N-central, these inequalities give $2\varphi(q) \leq \varphi(p) \leq \varphi(q)$, hence $\varphi(p) = \varphi(q) = 0$.

The unitary $h = w_1 a$ is B-Haar: every non-zero power is an alternating reduced word in centered P- and Q-letters, so $E_B(h^n) = 0$ for $n \neq 0$. For $m \neq n$,

$$h^n e_B h^{-n} h^m e_B h^{-m} = h^n e_B h^{m-n} e_B h^{-m} = h^n E_B(h^{m-n}) e_B h^{-m} = 0.$$

The projections $h^n e_B h^{-n}$ are pairwise orthogonal and, by centrality, have the same φ-value; hence $\varphi(e_B) = 0$. This contradicts $1 = \varphi(e_B + p + q)$. By DKEP Theorem 6.2 this is precisely proper proximality relative to X_B (and, in the factor case used below, agrees with the normal-bidual criterion from Lemma 8.5). $\square$

Application to graph products

Fix $v \in V$ and put $L_v = \mathrm{Lk}_\Gamma(v)$. The graph-product reduced-word model, or [3], gives the standard amalgamated free product decomposition

$$M \cong M_{V\setminus\{v\}} *_{M_{L_v}} (M_{L_v} \overline{\otimes} M_v). \tag{8}$$

Let $C(v) = \{r \neq v : r \not\sim_\Gamma v\}$. Since Γ^c is connected, $C(v) \neq \emptyset$. If $C(v)$ contains distinct r, t, choose trace-zero unitaries u_r, u_t and set $w_1 = u_r, w_2 = u_t$. Then $w_i \notin M_{L_v}$, and the reduced-word expectation onto M_{L_v} kills w_i and $w_1^* w_2$. If $C(v) = \{r\}$, connectedness of Γ^c and $|V| \geq 3$ give $s \in L_v$ with $s \not\sim_\Gamma r$; otherwise $\{v, r\}$ would be a component of Γ^c. Set $w_1 = u_r, w_2 = u_s u_r$. Here $E_{M_{L_v}}(w_1) = 0$, $E_{M_{L_v}}(w_2) = u_s E_{M_{L_v}}(u_r) = 0$, and $w_1^* w_2 = u_r^* u_s u_r$ is a reduced word containing the outside vertex r, so its expectation onto M_{L_v} is 0. Thus in all cases

$$E_{M_{L_v}}(w_i) = 0, \qquad E_{M_{L_v}}(w_1^* w_2) = 0.$$

A trace-zero unitary in M_v supplies the unitary a in (8), so Proposition 1 shows that M is properly proximal relative to $X_{M_{L_v}}$ for every v.

Enumerate $V = \{v_1, \ldots, v_m\}$. Since no vertex belongs to its own link, $\cap_i L_{v_i} = \emptyset$. Lemma 4 puts the associated palindromic product in $q_\emptyset \mathcal{A}_M q_\emptyset$. Here $M_\emptyset = \mathbb{C}$, $e_\mathbb{C}$ is the rank-one projection onto $\mathbb{C}\hat{1}$, and DKEP Example 3.4 gives $X_{M_\emptyset} = \mathbb{K}(L^2 M)$.

Finally, CDHJKN Theorem 2.4 implies that a join-irreducible graph product with $|V| \geq 3$ and trace-zero vertex unitaries is a diffuse full factor. A diffuse amenable finite factor is the hyperfinite II_1 factor and is not full, so M is non-amenable. It is separable because the graph is finite and the vertices are separable. Lemma 5, applied to the finite family $\{X_{M_{L_v}} : v \in V\}$, proves that M is properly proximal.

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